ADC Peptide Mapper

Version 0.5 — User Manual & Vignette

In-silico Tryptic Digest & Skyline Transition List Generator

Purpose	Step-by-step guide for running in-silico ADC tryptic digestion, uniqueness checking, and Skyline-compatible transition list export.
Audience	Bioanalytical scientists, proteomics researchers, and LC-MS/MS method developers working with Antibody-Drug Conjugates.
Version	v0.5 — 4-tab build (Input & Setup | Modifications | Peptide Results | Transition List)
Requirements	R ≥ 4.3, RStudio (recommended), internet access for first-run database build.

For research use only. Monoisotopic masses throughout. Background databases sourced from UniProt Swiss-Prot reviewed proteomes.

Table of Contents

1. Quick Start	3
2. Installation & Setup	3
3. Application Overview	4
4. Tab 1 — Input & Setup	5
5. Tab 2 — Modifications	6
6. Tab 3 — Peptide Results	8
7. Tab 4 — Transition List	9
8. Output File Reference	10
9. Scientific Background & Assumptions	11
10. Troubleshooting	12

1. Quick Start

The fastest path from FASTA to Skyline-ready transition list is five steps:

Step	Action	Where
1	Install R packages (one-time)	R console
2	Run build_background_db.R to download & cache proteome databases (one-time, ~10 min)	R console
3	Launch the app: shiny::runApp("app.R")	R console
4	Upload your ADC FASTA → select background species → click Run Digest	Tab 1
5	Configure instrument mode → Generate Transition List → Download Skyline CSV	Tab 4

2. Installation & Setup

2.1 R Package Installation

Open an R console or RStudio and run the following once:

```
install.packages(c(
  "shiny", "bs4Dash", "DT", "data.table", "openxlsx",
  "httr2", "stringr", "dplyr", "shinycssloaders", "shinyjs"
))
```

2.2 Build Background Databases (One-Time)

The app ships without the large proteome database files to keep the download small. Run the bundled script once to download and cache them. An internet connection is required. This takes approximately 10 minutes.

```
setwd('path/to/adc_peptide_mapper')  # your unzipped folder
source('build_background_db.R')
```

This creates three files in the **data/** folder:

- **bg_human.rds** — UniProt Swiss-Prot human reviewed (~20,442 proteins, 23 MB)
- **bg_monkey.rds** — Cynomolgus monkey (~1,177 proteins, 0.7 MB)
- **bg_rat.rds** — Rat (~8,231 proteins, 8.1 MB)

2.3 Launch the App

```
shiny::runApp('app.R')
```

Tip: In RStudio, open app.R and click the Run App button. The app opens in your browser at http://127.0.0.1:XXXX.

3. Application Overview

ADC Peptide Mapper is a browser-based R Shiny application with a dark navy sidebar and four sequential tabs. Work through the tabs left-to-right for a complete analysis.

Tab	Name	Purpose
1	**Input & Setup**	Upload ADC FASTA, select background proteome, configure digest parameters, run digest.
2	**Modifications**	Configure fixed, variable, special, and custom post-translational modifications.
3	**Peptide Results**	Browse, filter, and export the full modified peptide table with uniqueness flags.
4	**Transition List**	Generate and download Skyline-compatible MRM or HR transition lists.

Workflow summary: The app performs in-silico trypsin digestion of your ADC sequence(s), enumerates all selected modification combinations, computes monoisotopic masses, checks each peptide's bare sequence against a pre-built background proteome digest, and flags peptides unique to your ADC. The transition list tab then calculates precursor m/z, fragment ion series (b/y/a), and empirical collision energies for direct import into Skyline.

4. Tab 1 — Input & Setup

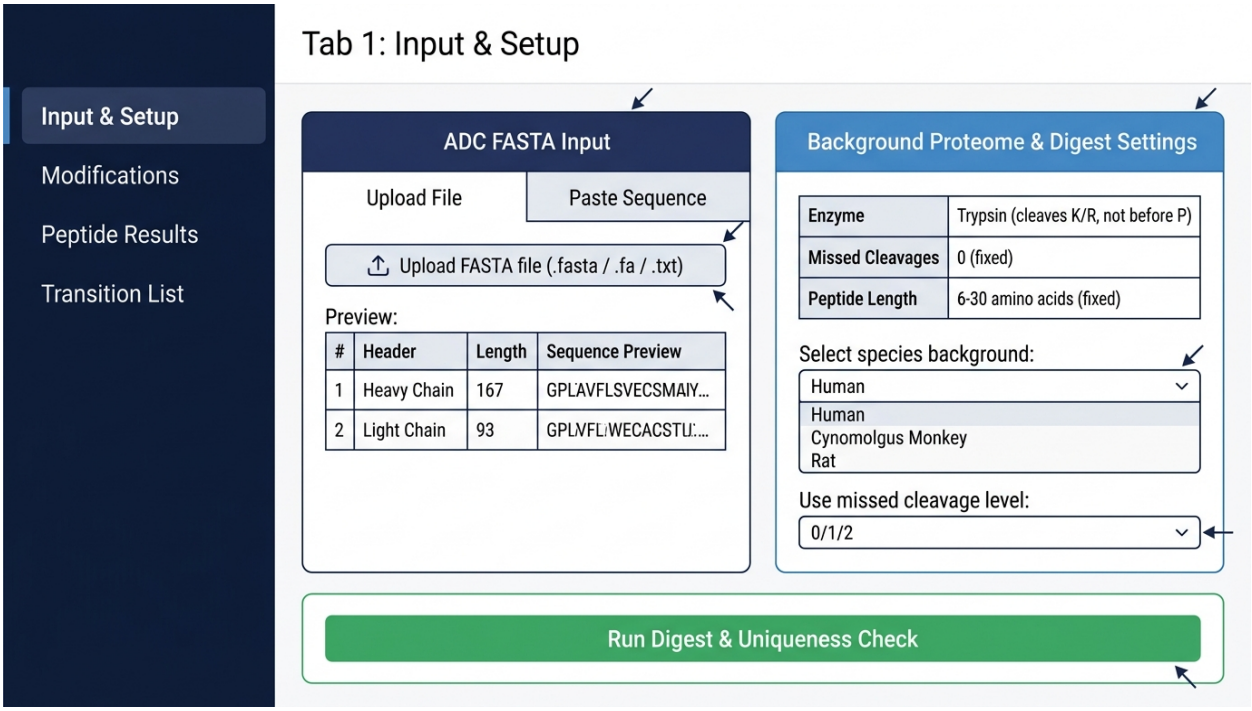

Figure 1. Tab 1: Input & Setup interface.

4.1 ADC FASTA Input

Provide your ADC sequence in FASTA format using one of two methods:

- **Upload File** — Click Browse and select a .fasta, .fa, or .txt file. Multi-chain FASTAs (e.g., Heavy Chain + Light Chain) are fully supported.
- **Paste Sequence** — Type or paste directly into the text area. A Trastuzumab HC+LC demo sequence is pre-loaded for immediate testing.

After input, the app automatically parses each **>header** entry as a separate chain and displays a preview table showing chain number, header name, sequence length (AA), and the first 60 residues.

4.2 Background Proteome Settings

Parameter	Details
Select species background	Choose from pre-built databases: Human (UniProt Swiss-Prot, ~20,442 proteins), Cynomolgus Monkey (~1,177 proteins), or Rat (~8,231 proteins). Select 'Custom FASTA' to upload your own background.
Use missed cleavage level	For bundled databases, all missed cleavage levels (MC=0, 1, 2) are pre-computed. Select 0 for strict uniqueness (recommended), 1 or 2 for broader background coverage.
Custom FASTA upload	Visible only when 'Custom FASTA' is selected. Upload any FASTA file as background. Also select the missed cleavage level and click Build Custom Background.

4.3 Fixed Digest Parameters

The following parameters are fixed and cannot be changed:

Parameter	Details
Enzyme	Trypsin — cleaves after K or R, but not when followed by P (KP/RP rule).
Missed cleavages	0 — no missed cleavages for ADC peptides.
Peptide length	6–30 amino acids. Shorter or longer peptides are discarded.
Mass type	Monoisotopic masses throughout (not average masses).

4.4 Running the Digest

Click the **Run Digest & Uniqueness Check** button (green, bottom of page). A progress indicator appears while the app digests your ADC sequence(s), applies all selected modifications, and compares each peptide against the background. Results appear in Tab 3 when complete.

5. Tab 2 — Modifications

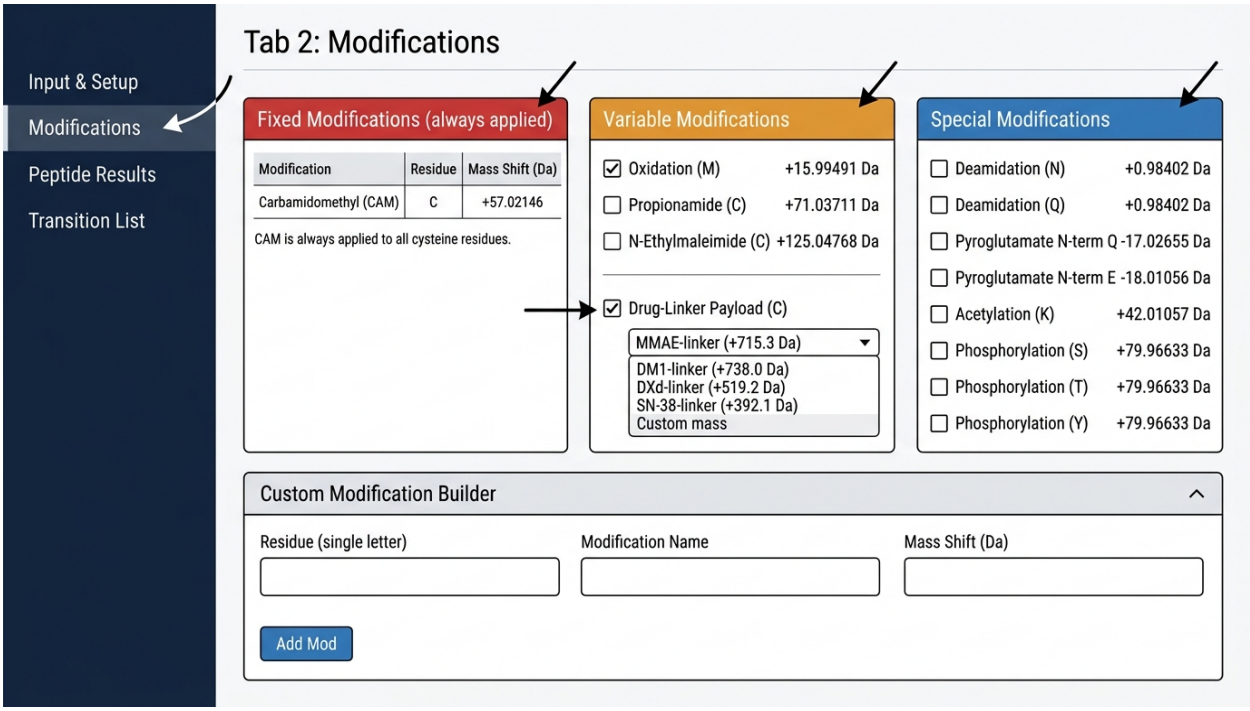

Figure 2. Tab 2: Modifications interface.

Configure modifications before running the digest. Changes take effect on the next **Run Digest** click. The tab is divided into four panels:

5.1 Fixed Modifications (always applied)

Modification	Residue	Mass Shift (Da)	Notation
Carbamidomethylation (CAM)	C (Cysteine)	+57.02146	C[CAM]

CAM is applied to every cysteine residue in every peptide. It cannot be disabled.

5.2 Variable Modifications (checkboxes)

Each enabled variable modification generates additional peptide variants. All combinations of enabled mods are enumerated (capped at 4 variable sites per peptide to prevent combinatorial explosion).

Modification	Residue	Mass Shift (Da)	Default	Notes
Oxidation	M (Methionine)	+15.99491	ON	Most common in-solution artifact.
Propionamide	C (Cysteine)	+71.03711	OFF	Acrylamide adduct from gel electrophoresis.
N-Ethylmaleimide (NEM)	C (Cysteine)	+125.04768	OFF	Cysteine-reactive labelling reagent.

Drug-Linker Payload	C (Cysteine)	See below	OFF	Enable to select ADC payload.

5.3 Drug-Linker Payload Options

When **Drug-Linker Payload (C)** is checked, select the payload from the dropdown:

Payload	Mass Shift (Da)	Target Residue
MMAE-linker	+715.3	C
DM1-linker	+738.0	C
DXd-linker	+519.2	C
SN-38-linker	+392.1	C
Custom mass	User-defined numeric input	C

Drug-linker payloads are applied to cysteine residues. When a payload is selected, it is treated as a variable modification — peptides with and without the payload are both generated.

5.4 Special Modifications

Modification	Residue	Mass Shift (Da)	Condition
Deamidation	N (Asparagine)	+0.98402	Any position
Deamidation	Q (Glutamine)	+0.98402	Any position
Pyroglutamate	Q (Glutamine)	–17.02655	N-terminal only
Pyroglutamate	E (Glutamic acid)	–18.01056	N-terminal only
Acetylation	K (Lysine)	+42.01057	Any position
Phosphorylation	S (Serine)	+79.96633	Any position
Phosphorylation	T (Threonine)	+79.96633	Any position
Phosphorylation	Y (Tyrosine)	+79.96633	Any position

5.5 Custom Modification Builder

Add any modification not listed above using the Custom Modification Builder panel:

- **Residue** — Single-letter amino acid code (e.g., K, R, W).
- **Modification Name** — A descriptive label (e.g., Ubiquitination).
- **Mass Shift (Da)** — Monoisotopic mass delta (positive or negative).
- Click **Add Mod** to add it to the active list. Remove entries with the delete button in the table.

6. Tab 3 — Peptide Results

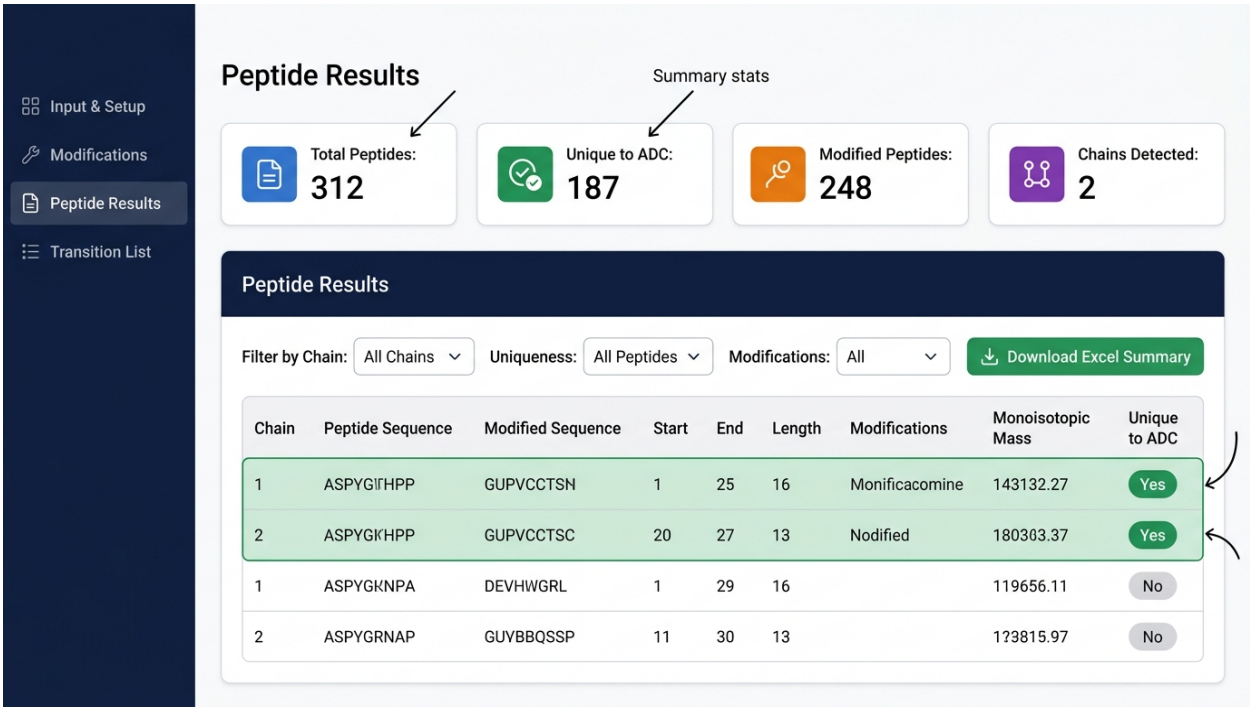

Figure 3. Tab 3: Peptide Results with summary cards and filterable table.

6.1 Summary Cards

Four value boxes at the top of the tab provide an at-a-glance digest summary:

Card	Description
Total Peptides	Total number of peptide variants generated (all chains, all mod combinations).
Unique to ADC	Peptides whose bare sequence is absent from the selected background proteome digest.
Modified Peptides	Peptide variants carrying at least one variable or special modification.
Chains Detected	Number of FASTA chains (headers) parsed from the input.

6.2 Filters

Parameter	Details
Filter by Chain	Show peptides from a specific chain (e.g., Heavy Chain, Light Chain) or All Chains.
Uniqueness	All Peptides / Unique to ADC only / Non-unique only.
Modifications	All / Modified only (at least one variable mod) / Unmodified only (CAM only).

6.3 Results Table Columns

Column	Description
Chain	Chain label from the FASTA header (e.g., Trastuzumab_HeavyChain).
Peptide Sequence	Bare amino acid sequence (no modification annotations).
Modified Sequence	Sequence with inline modification tags, e.g. C[CAM], M[Ox].
Start / End	1-based position of the peptide within the parent protein sequence.
Length	Number of amino acid residues.
Modifications	Semicolon-separated list of applied modifications with position, e.g. Oxidation@M12.
Monoisotopic Mass	Neutral monoisotopic mass (Da) of the modified peptide.
Unique to ADC	Green 'Yes' badge if the bare sequence is absent from the background; grey 'No' if found.

6.4 Export

Click **Download Excel Summary** to export the full (filtered) peptide table as an .xlsx file with formatted headers. The download respects the current filter state.

7. Tab 4 — Transition List

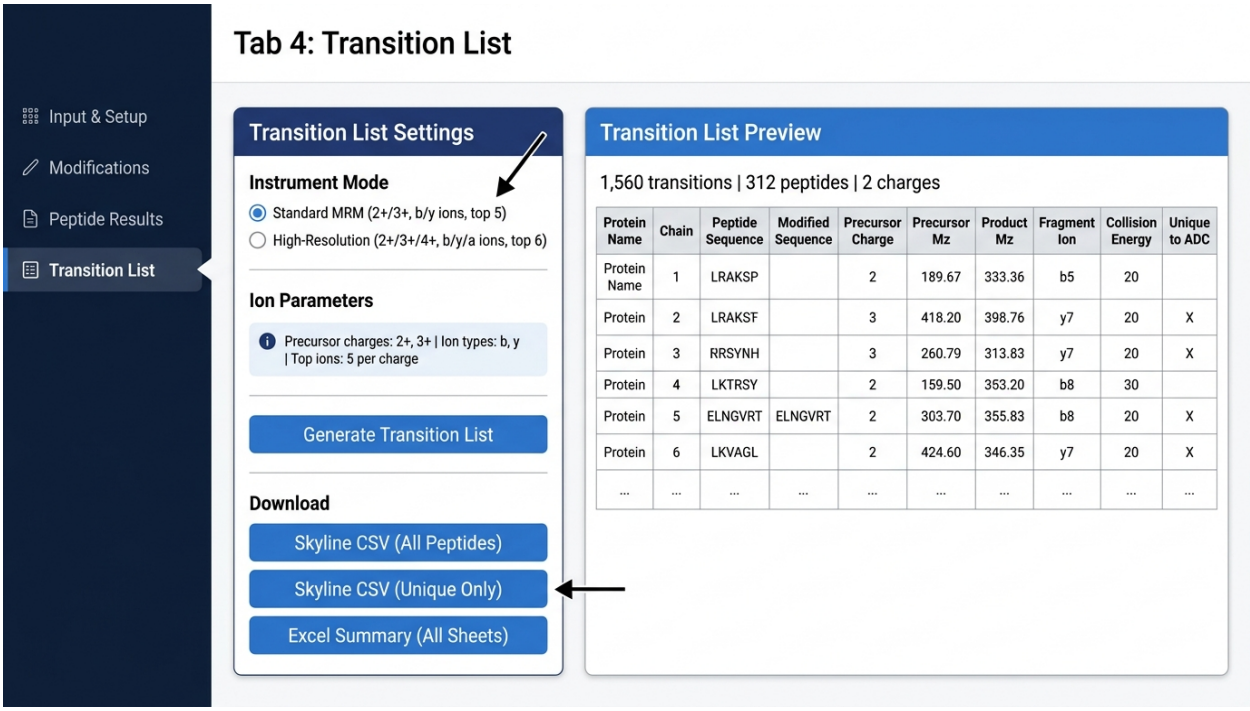

Figure 4. Tab 4: Transition List settings and preview.

7.1 Instrument Mode

Mode	Precursor Charges	Ion Types	Ions per Charge	Use Case
Standard MRM	2+, 3+	b, y	Top 5	Triple-quadrupole MRM methods (e.g., Sciex QTRAP, Waters Xevo).
High-Resolution	2+, 3+, 4+	b, y, a	Top 6	High-resolution instruments (e.g., Sciex TripleTOF, Thermo Orbitrap).

7.2 Ion Calculation Details

Parameter	Details
Precursor m/z	(Neutral mass + z × 1.007276) / z, where z is the precursor charge state.
b-ions	N-terminal fragment series. b_i = sum of residue masses [1..i] + 1.007276 Da (proton).
y-ions	C-terminal fragment series. y_i = sum of residue masses [(n–i+1)..n] + 18.01056 (water) + 1.007276 Da.
a-ions (HR only)	a_i = b_i – 27.99491 Da (loss of CO). Generated only in High-Resolution mode.
Product ion charge	All fragment ions are singly charged (z = 1).

Minimum ion size	Single-residue terminal ions (b1, y1, a1) are excluded.
Top-N selection	Ions are ranked by fragment number (larger fragments preferred) and the top N per ion type are retained.
Collision Energy (CE)	Sciex empirical formula: CE = 0.0448 × (m/z) – 2.0 for 2+; CE = 0.0533 × (m/z) – 2.0 for 3+; CE = 0.0580 × (m/z) – 2.0 for 4+. Minimum CE is 10 eV. Values can be overridden in Skyline after import.

7.3 Generating and Downloading

- Click **Generate Transition List** to compute transitions for all peptides in the results table.
- A preview table appears on the right showing all transitions with m/z values and CE.
- **Skyline CSV (All Peptides)** — exports every transition for direct import into Skyline.
- **Skyline CSV (Unique Only)** — exports only transitions for peptides flagged as unique to the ADC.
- **Excel Summary (All Sheets)** — exports a multi-sheet workbook: Transitions, Peptide Summary, and Modifications.

8. Output File Reference

8.1 Skyline CSV Format

The exported CSV is formatted for direct import into Skyline via *File → Import → Transition List*. Column definitions:

Column	Type	Description
Protein Name	Text	FASTA header of the parent protein/chain.
Peptide Sequence	Text	Bare amino acid sequence.
Modified Sequence	Text	Sequence with inline mod tags (e.g., C[CAM], M[Ox]).
Precursor Charge	Integer	Charge state of the precursor ion (2, 3, or 4).
Precursor Mz	Decimal	Precursor m/z (5 decimal places).
Product Charge	Integer	Always 1 (singly charged fragments).
Product Mz	Decimal	Fragment ion m/z (5 decimal places).
Fragment Ion	Text	Ion label, e.g. b5, y7, a3.
Collision Energy	Decimal	Empirical CE in eV (1 decimal place).
Unique To ADC	Boolean	TRUE if peptide is absent from background proteome.
Chain	Text	Chain label from FASTA header.
Modifications	Text	Semicolon-separated modification list with positions.

8.2 Excel Summary Workbook

Sheet	Contents
Peptide Results	Full modified peptide table matching Tab 3 display.
Transition List	Full transition list matching Tab 4 display.
Modifications	Summary of all active modifications applied in the run.

9. Scientific Background & Assumptions

9.1 Trypsin Digestion Rules

Trypsin cleaves peptide bonds on the C-terminal side of lysine (K) and arginine (R) residues. Cleavage is suppressed when the next residue is proline (P) — the KP and RP rules — because the cyclic structure of proline sterically hinders trypsin access. This app enforces the KP/RP rule strictly. Missed cleavages for the ADC are fixed at 0.

9.2 Monoisotopic Masses

All mass calculations use monoisotopic masses (most abundant isotope of each element: ^{1}H, ^{12}C, ^{1}■N, ^{1}■O, ^{32}S). This is the standard for high-resolution MS and MRM method development. Average masses are not used.

9.3 Uniqueness Definition

A peptide is flagged as **Unique to ADC** if its bare (unmodified) amino acid sequence does not appear in the background proteome digest at the selected missed cleavage level. Modifications are not considered during uniqueness comparison — only the sequence. This is intentional: a peptide unique at the sequence level remains unique regardless of which modifications are present.

9.4 Background Proteome

Background databases are built from UniProt Swiss-Prot reviewed proteomes — the highest-confidence, manually curated subset of UniProt. Peptide counts per species at MC=0:

Species	Proteins	Peptides (MC=0)	Peptides (MC=1)	Peptides (MC=2)
Human (Homo sapiens)	20,442	548,195	1,353,560	2,083,004
Cynomolgus Monkey	1,177	26,986	67,662	104,894
Rat (Rattus norvegicus)	8,231	210,549	515,946	788,405

9.5 Key Assumptions

#	Assumption
1	Monoisotopic masses used throughout.
2	Trypsin KP/RP rule strictly enforced.
3	CAM on cysteine is always a fixed modification and cannot be disabled.
4	Uniqueness check uses bare (unmodified) sequences only.
5	All product ions are singly charged (z = 1).
6	Single-residue terminal ions (b1, y1, a1) are excluded from transition lists.
7	CE formula is Sciex-style empirical; override in Skyline for other vendors.
8	Variable mod combinations are capped at 4 sites per peptide to prevent combinatorial explosion.

10. Troubleshooting

Problem	Likely Cause	Solution
App fails to start	Missing R packages.	Re-run the install.packages() command from Section 2.1.
'No bundled databases found' warning	build_background_db.R has not been run, or data/ folder is missing.	Run source('build_background_db.R') from the app directory.
0 peptides generated	FASTA format error, or all peptides fall outside 6–30 AA length filter.	Check FASTA has valid >header lines. Verify sequence contains K/R residues for trypsin cleavage.
All peptides show Unique = No	Wrong background species selected, or MC level too high (more background peptides).	Confirm species matches your study. Try MC=0 for strictest uniqueness.
Transition list is empty	Peptide results table is empty, or Generate Transition List not clicked.	Run digest first (Tab 1), then click Generate Transition List (Tab 4).
Skyline import fails	Column name mismatch or encoding issue.	Ensure Skyline is set to import 'Transition List' format. Use the unmodified CSV download.
build_background_db.R fails	No internet connection, or UniProt API timeout.	Check internet access. Re-run the script — it resumes from cached files if partially complete.
Drug payload mass not applied	Drug-Linker Payload checkbox not enabled.	In Tab 2, check the 'Drug-Linker Payload (C)' checkbox, then select the payload and re-run digest.

ADC Peptide Mapper v0.5 — For research use only. Monoisotopic masses | Trypsin KP/RP rule | UniProt Swiss-Prot backgrounds.